11.3	–1.68	2.8224
13	0.02	0.0004
15	2.02	4.0804
12	–0.98	0.9604
9	–3.98	15.8404
***n* =10**		**71.136**

$$\text{Standard deviation } \sigma = \sqrt{\frac{\sum d^2}{n}}$$

$$= \sqrt{\frac{71.136}{10}}$$

$$= 2.67$$

NOTE

If the mean value is not an integer, the calculation is difficult. In such a case we use the alternative formula for the calculation.

Example 6.7

Find the standard deviation of the first '*n*' natural numbers.

Solution:

The first *n* natural numbers are 1, 2, 3,…, *n*. The sum and the sum of squares of these *n* numbers are

$$\sum x_i = 1+2+3+.....+n = \frac{n(n+1)}{2}$$

$$\sum x^2_i = 1^2 + 2^2 + 3^2 + ...n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Mean } \bar{x} = \frac{1}{n}\sum x_i = \frac{n(n+1)}{2n} = \frac{(n+1)}{2}$$

$$\frac{\sum x_i^2}{n} = \frac{(n+1)(2n+1)}{6}$$

$$\text{Standard deviation, } \sigma = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

$$= \sqrt{\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}}$$

$$= \sqrt{\frac{2(n+1)(2n+1) - 3(n+1)^2}{12}}$$

$$= \sqrt{\frac{(n+1)[2(2n+1) - 3(n+1)]}{12}}$$

$$= \sqrt{\frac{(n+1)(n-1)}{12}} = \sqrt{\frac{(n^2-1)}{12}}$$

$$\sigma = \sqrt{\frac{(n^2-1)}{12}}$$

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